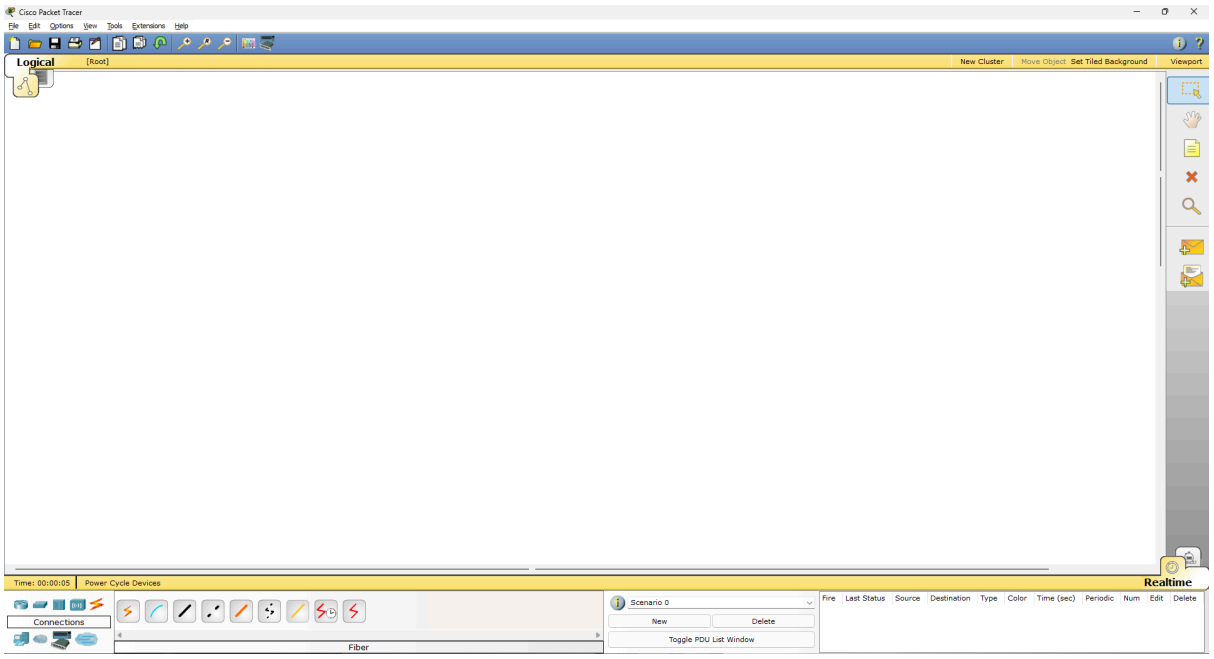
PRACTICALS

Computer Networks Laboratory

Practical 8.

Aim: To implement the RIP routing protocol in the given network having atleast 4 routers.

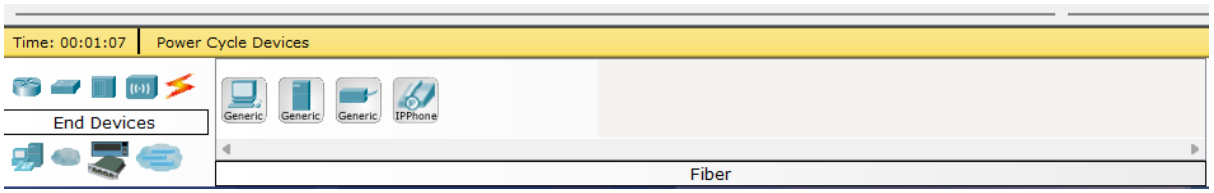
Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**



[FIG 1]

Step 2: Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**
There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 3 - PCs
2. 1 - Network Switch
3. 3 - Connecting wires/ cables



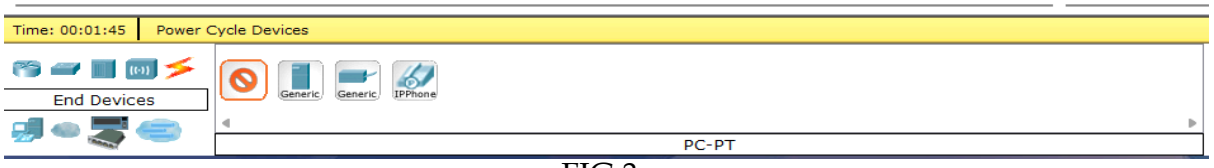


FIG 3

Step 4: Creating Network (Topology) Using selected devices as shown in **FIG 4**

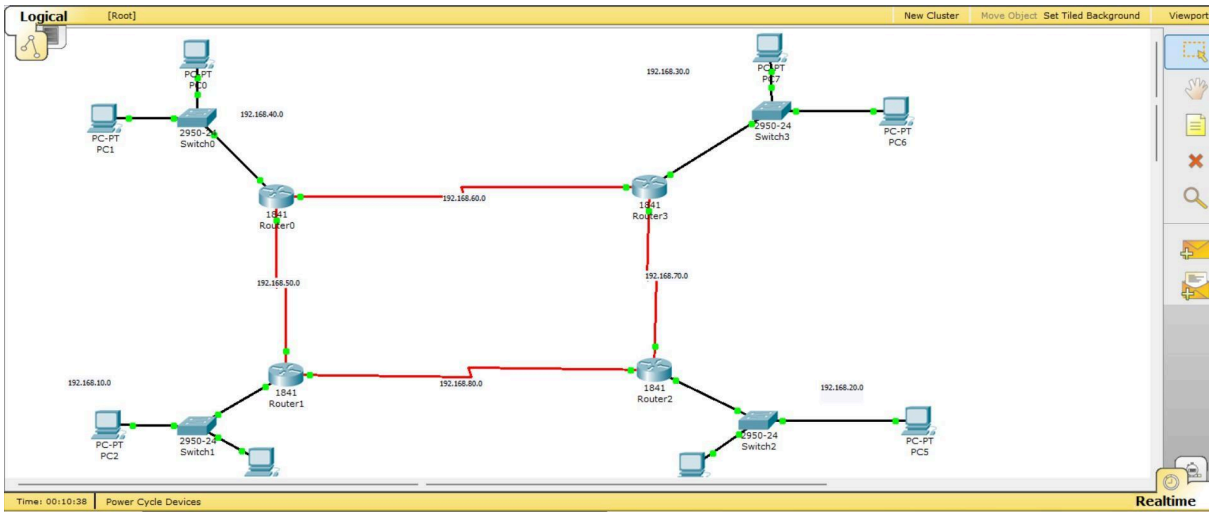


FIG 4

Step 5: Configuring The Setup with Network (static method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

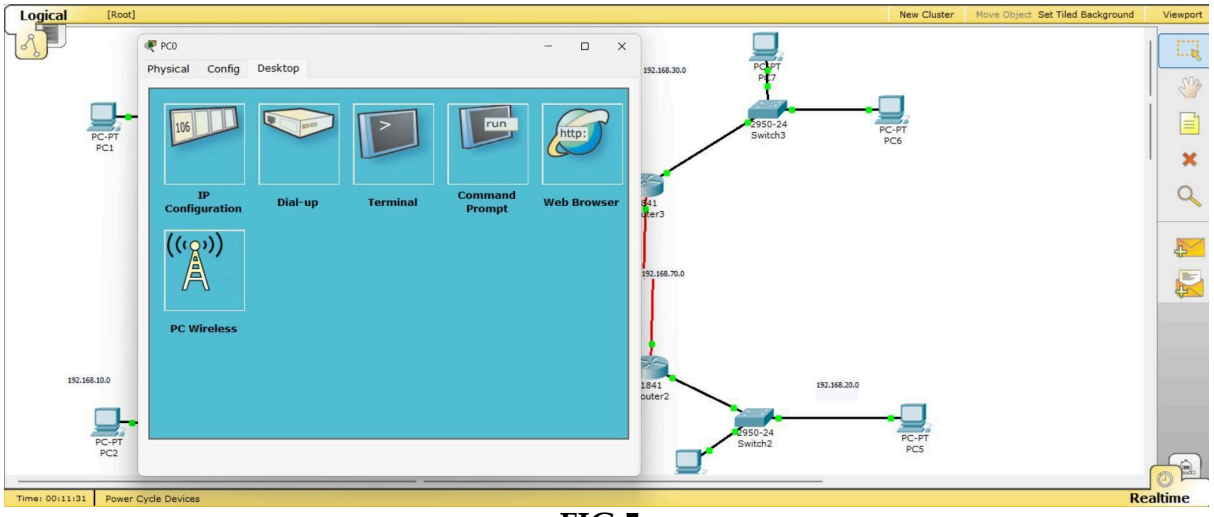


FIG 5

1. Select Or click on Static
2. Enter values as follows
3. Close the dialog box
4. Do the same process for all of the PCs

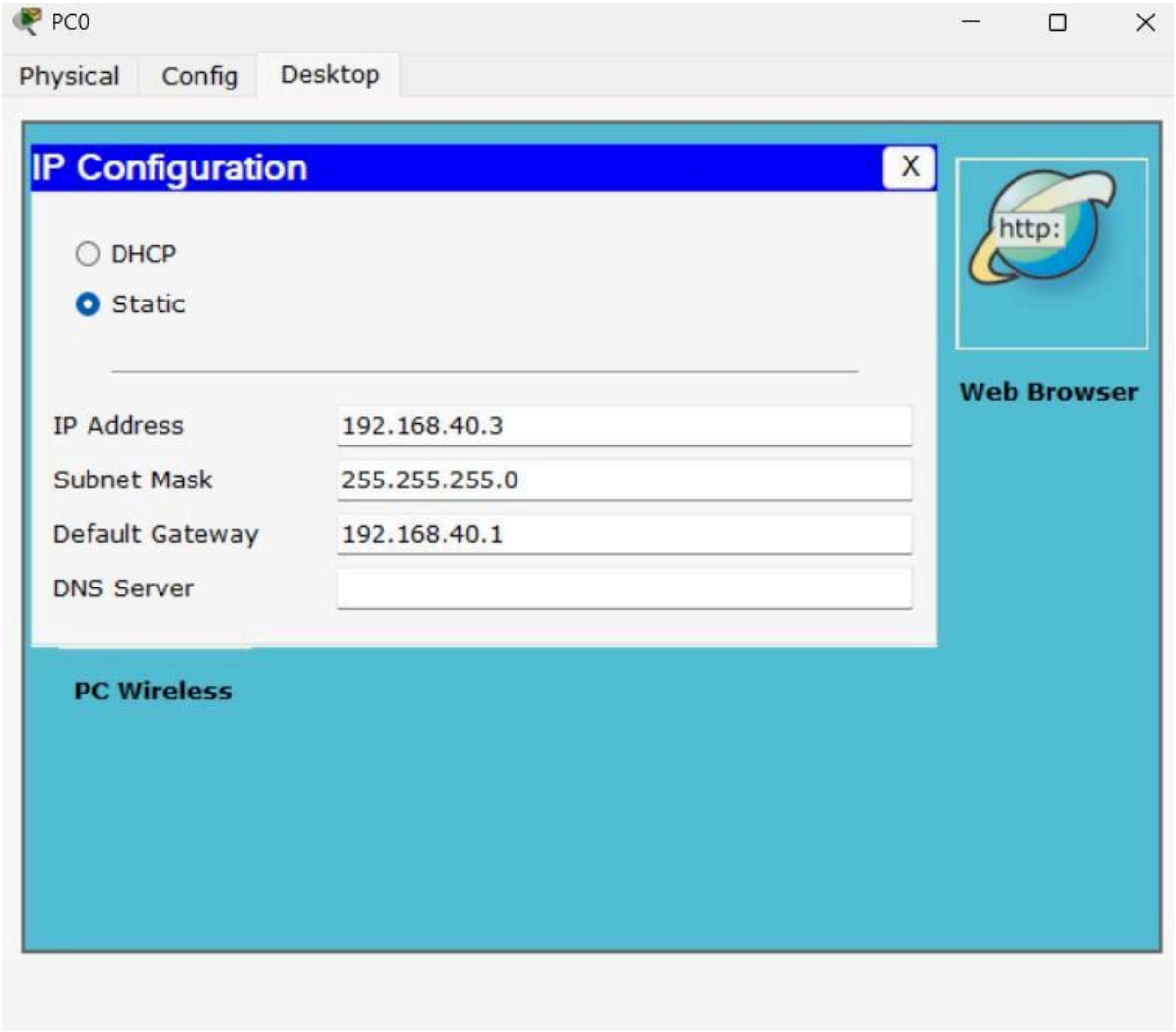


FIG 6

Note:- IP for all pcs are as follows (with subnet mask) Common Subnet mask for all Pcs:- **255.255.255.0**
PC 0,1:- 192.168.40.2, 192.168.40.3
PC 2,3,:- 192.168.10.2, 192.168.10.3,
PC 4,5:- 192.168.20.2, 192.168.20.3
PC 6,7:- 192.168.30.2, 192.168.30.3

Step 6: Now Configuring Routers with following steps:

1. Click on Router
2. Click on Config tab
3. Click on FastEathernet0/0 for setup router and network switch
4. Click on serial0/0/0 &serial0/0/0 for setup routers

Step 7: setup the values given in below **FIG 7** for FastEathernet 0/0

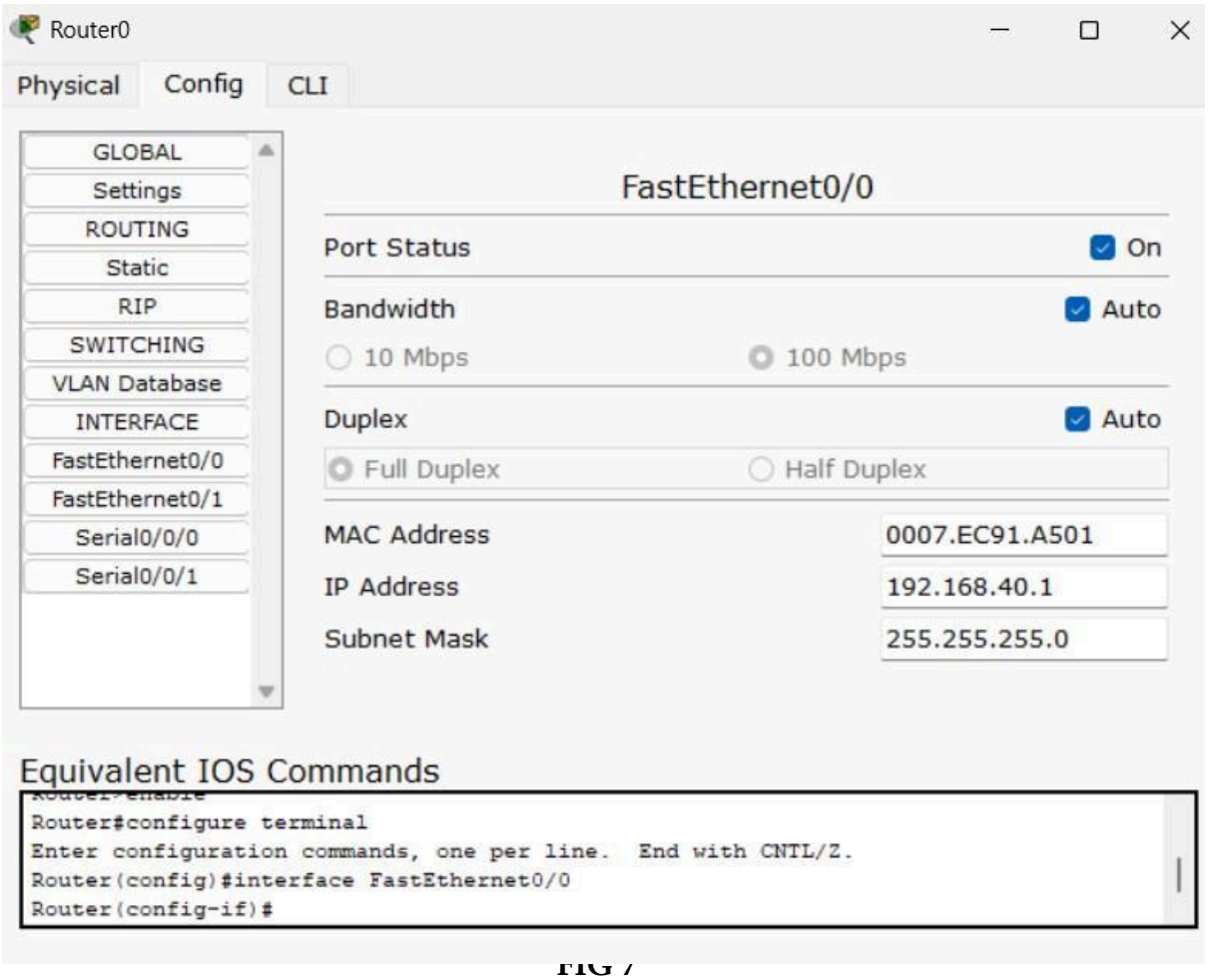
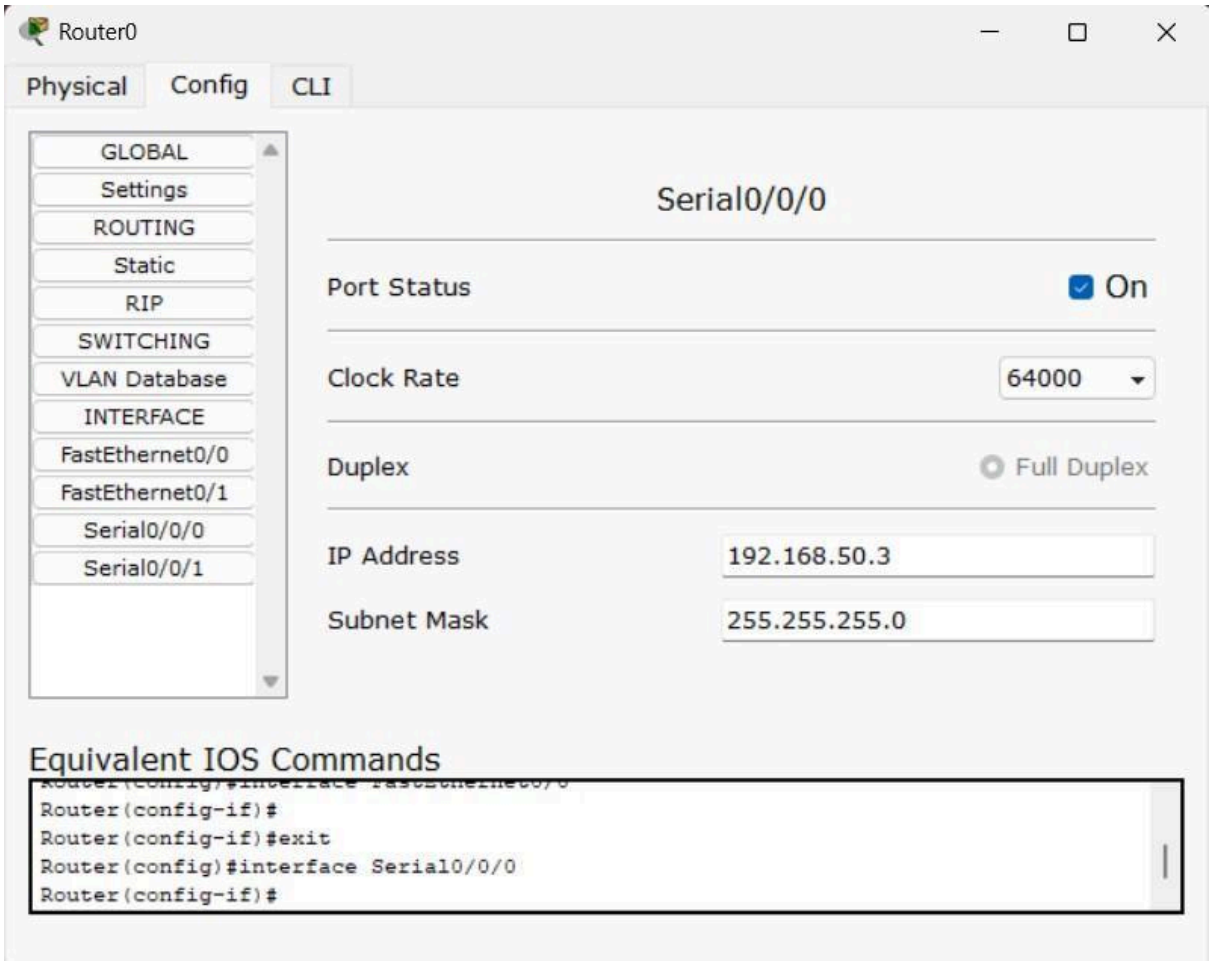


FIG 7

Step 8: setup the values given in below **FIG 8** for serial0/0/0 & **FIG 9** for serial0/0/1



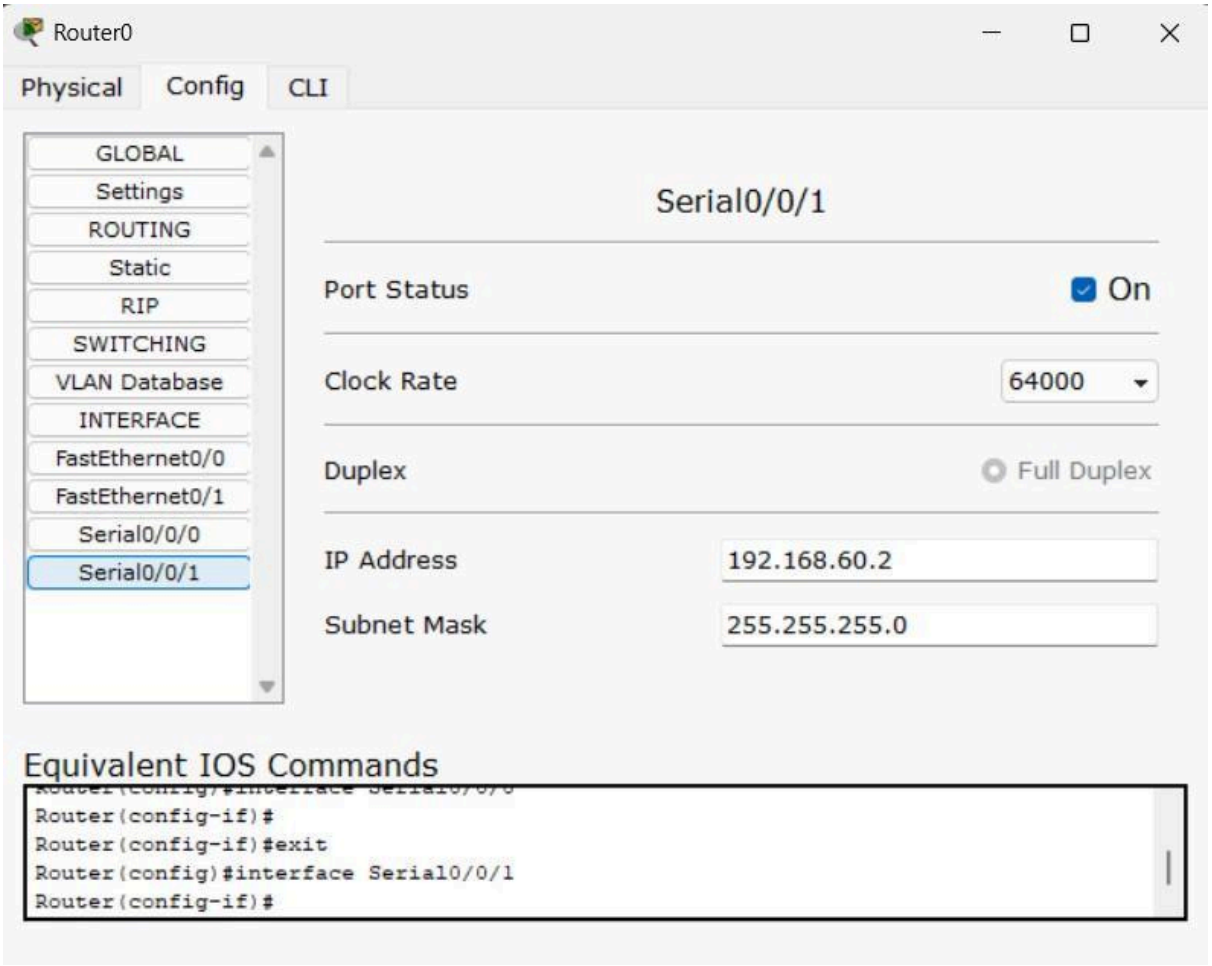


FIG 9

Note:- For other router refer the following table

Router 0

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.40.1	192.168.50.3	192.168.60.2
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 1

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.10.1	192.168.50.2	192.168.80.2
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 2

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.20.1	192.168.70.3	192.168.80.3
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 3

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.30.1	192.168.70.2	192.168.60.3
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Step 9: Go to Router and click on Config and goto RIP and add network as shown in **FIG 10** for router 0

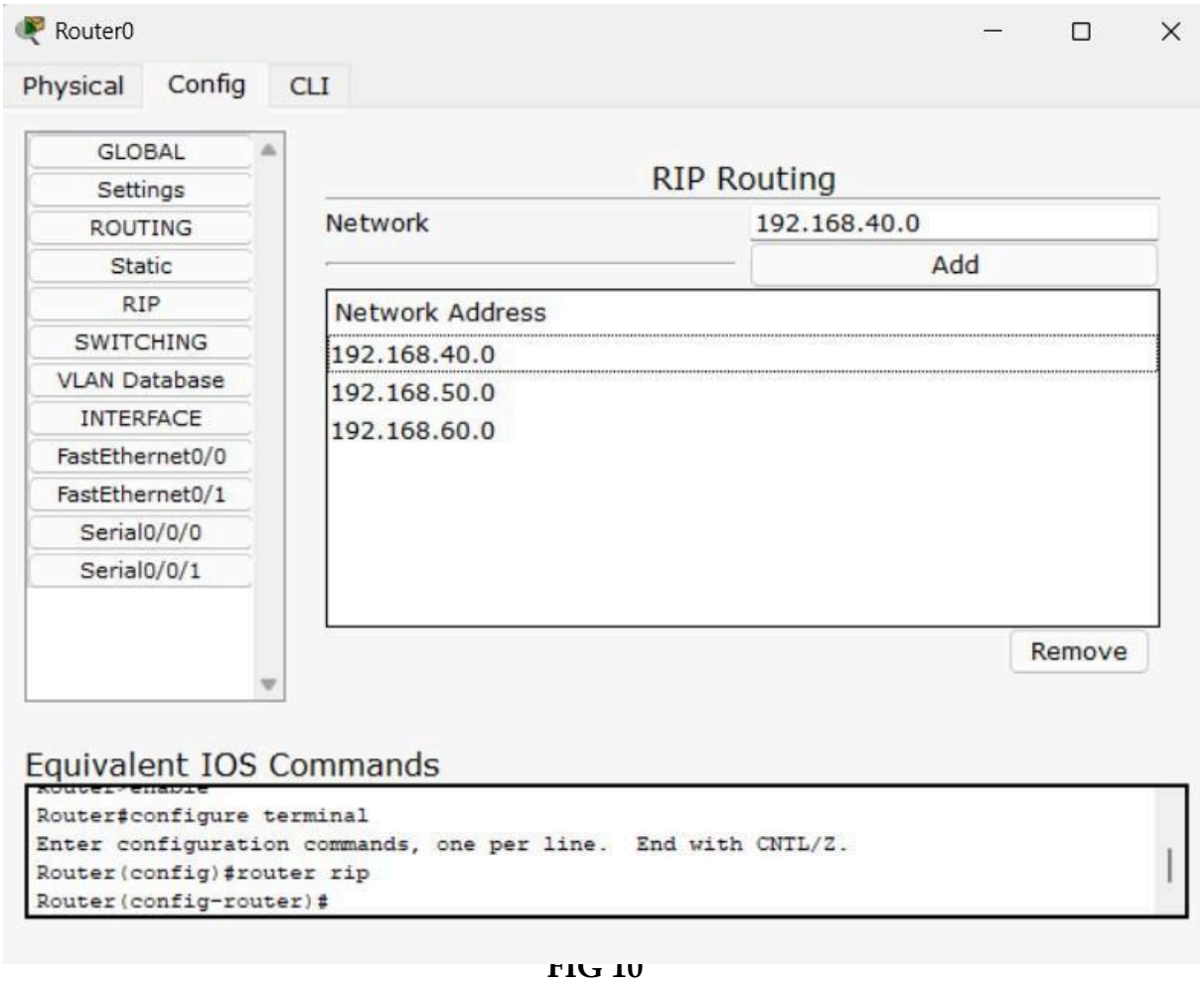


FIG 10

Note: Follow the following table for all routers

Router 0
192.168.40.0 192.168.50.0 192.168.60.0

Router 1
192.168.10.0 192.168.50.0 192.168.80.0

Router 2
192.168.20.0 192.168.70.0 192.168.80.0

Router 3
192.168.30.0 192.168.60.0 192.168.70.0

Step 10: Try to send Packets Through-out the network

At first attempt it will failed (to share outside the network) but in second attempt of sending packets it will success as shown in **FIG 11**

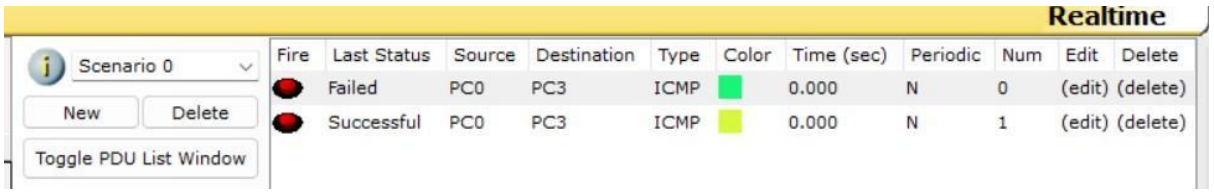


FIG 11